

ASSESSMENT TOOL - 3

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1) The concept of image enhancement in the spatial domain.

The image enhancement is the process of improving the quality of an image so it becomes easier to view. In the spatial domain,

1. Basic grey-level Transformations.

- i) Image negative - Highlights details in dark regions.
- ii) Log Transformation - Brightens dark images.
- iii) Power-Law : Adjusts image brightness and contrast.

2) Histogram Processing Technique.

- i) Histogram Equalization: Improves contrast by spreading pixel intensity values evenly.
- ii) Histogram stretching: Expands the intensity range to make the image clearer.

3) Spatial Filtering:

- Smoothing Filters: Remove noise and produce smoother pixels.
- Sharpening Filters: Enhances edges and fine details.

Conclusion

Image enhancement in the spatial domain improves the visual quality of captured images by directly modifying pixel values.

2) Difference between image smoothing and image sharpening.

Image smoothing	Image sharpening.
Remove noise & unwanted details	Enhances edges and fine details.
Produces a blurred or smoother image.	Produces a clearer and sharper image.
Uses low pass filters.	Uses high-pass filters.
Ex: mean filter.	Ex: Laplacian filter.

Working Principle of spatial smoothing.

Spatial smoothing filters work by replacing each pixel with the average or weighted average of its neighboring pixels.

Ex:

mean filter: Replaces each pixel with average blur.

Gaussian filter: uses weighted averaging to smooth.

Working Principle of Sharpening Filters.

Sharpening filters highlight edges by increasing the intensity difference b/w neighboring pixels.

Ex:

- Laplacian Filters: detects and enhances edge.
- Sobel Filter: Highlights horizontal and vertical edges.

~~When to Apply Each Technique.~~

- ~~Image Smoothing~~: Used when the image contains noise, grain, or unwanted variations, such as images.
- Image sharpening: used when the image appears blurred.

3) Image Enhancement in Frequency Domain.

Image enhancement in the frequency domain improves an image by changing its frequency components instead of directly changing pixel values.

Low - Pass Filter (LPF)

- Reduces noise.
- Produces a smooth image.
- Removes small unwanted details.

Low - Pass Filter	High - Pass Filter.
Removes noise Produces a smooth image Remove high - frequency used in image smoothing	Enhances edges. produces a sharp image. Remove Low frequency. used in image sharpening.

Conclusion :

Frequency domain enhancement improves image quality by filtering components. Low - Pass Filters are used for noise reduction and smoothing, while high - pass filters are used for edge enhancement and sharpening. The choice of filter on the image and the desired results.

Image segmentation.

1. Point detection.

- i) detect individual pixels that are different from their neighbours.
- ii) Advantage: simple and fast.
- iii) Limitations: sensitive to noise.
- iv) Application: detecting small defects.

2. Line detection

- i) detects horizontal, vertical or diagonal lines in an image.
- ii) Advantage: Finds straight lines clearly.
- iii) Limitations: Cannot detect curved shapes well.
- iv) Application: Road detection, document analysis.

3. Edge Detection.

- detects the boundaries of object by finding.
- Advantage: clearly show object boundaries.
- Limitations: can detect unwanted edges due to noise.
- Application: medical images, object detection.

5) 1. Edge detection

Edge detection finds boundaries of object by detecting sudden changes in brightness.

Working Principle:

- Compare the brightness of neighboring pixels.
- Large changes indicate an edge.

Ex: detecting the outline of car in traffic image.

2. Edge Linking.

Edge linking connects broken.

Working Principle:

- Joins edge pixels that are close to each other.
- Creates a complete object shapes.

Ex: Connecting broken road lines in satellite image.

3. Boundary Detection.

Boundary detection identifies the complete outer border.

Working Principle:

- uses linked edge to trace the object boundary.
- separates the object from the background.

Imposter:

Ex:

Detecting the boundary of a tumor in a medical image.